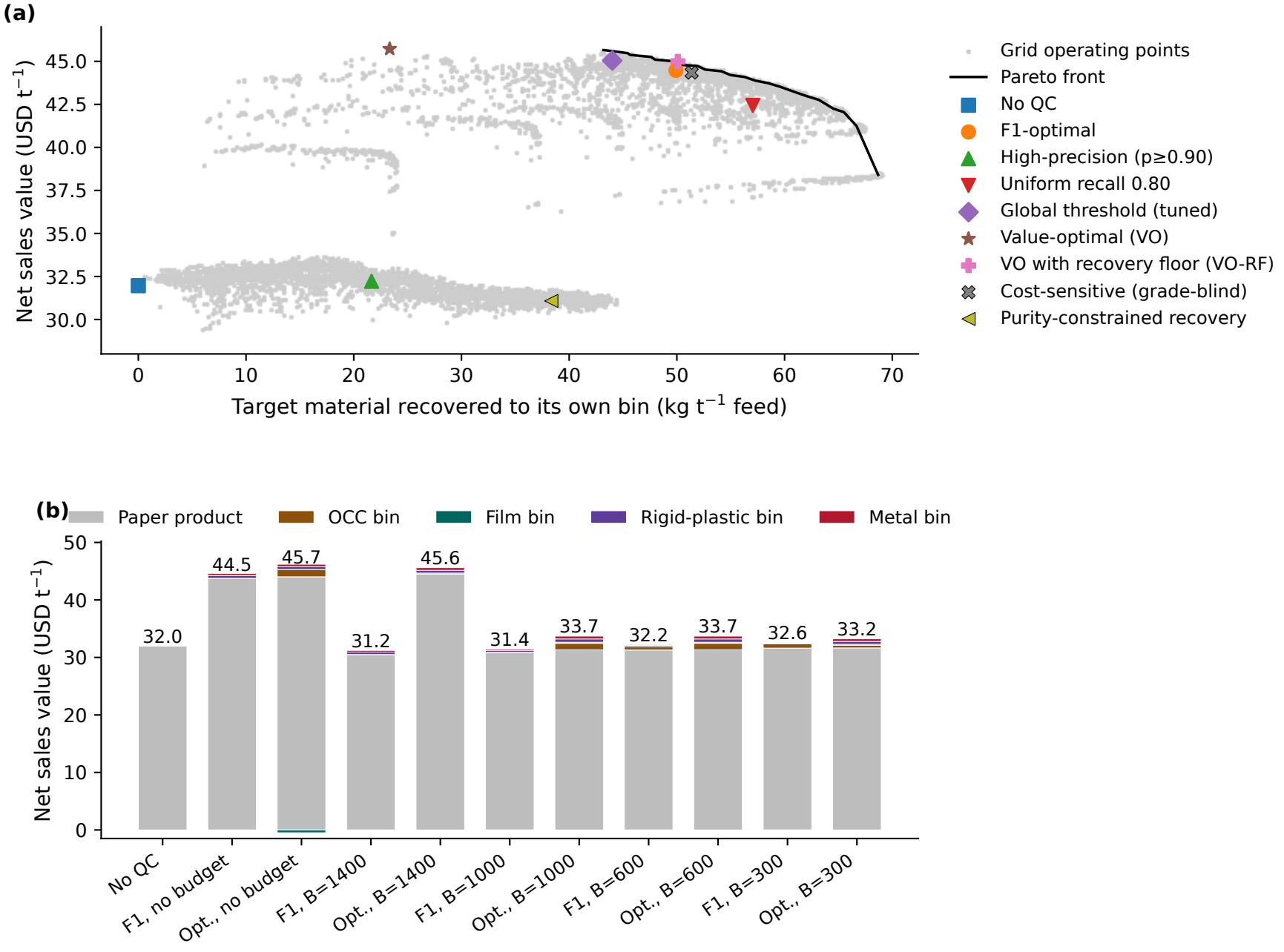
(a)
Net sales value (USD t^{-1})
45.0
42.5
40.0
37.5
35.0
32.5
30.0
0
10
20
30
40
50
60
70
Target material recovered to its own bin (kg t^{-1} feed)
Grid operating points
Pareto front
No QC
F1-optimal
High-precision (p≥0.90)
Uniform recall 0.80
Global threshold (tuned)
Value-optimal (VO)
VO with recovery floor (VO-RF)
Cost-sensitive (grade-blind)
Purity-constrained recovery
(b)
Paper product
OCC bin
Film bin
Rigid-plastic bin
Metal bin
Net sales value (USD t^{-1})
50
40
30
20
10
0
32.0
44.5
45.7
31.2
45.6
31.4
33.7
32.2
33.7
32.6
33.2
No QC
F1, no budget
Opt., no budget
F1, B=1400
Opt., B=1400
F1, B=1000
Opt., B=1000
F1, B=600
Opt., B=600
F1, B=300
Opt., B=300